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Education

VIT-AP University

B.Tech in Computer Science - Artificial Intelligence & Machine Learning CGPA: 8.05/10

2023 - Expected 2027

Amaravati, Andhra Pradesh

Technical Skills

Languages: Python, TypeScript, JavaScript, SQL, C++

ML / AI: PyTorch, TensorFlow/Keras, scikit-learn, Hugging Face (Transformers, PEFT, TRL), reinforcement learning (REINFORCE, PPO), RAG, FAISS, ReAct agents, DPO/QLoRA

Data / Audio: pandas, NumPy, SciPy, SQLite, librosa (STFT, mel, pYIN), dynamic time warping, pre-registered and matched-control experiments, bootstrap CIs

Engineering: FastAPI, Hono, React, REST APIs, Drizzle ORM, pytest, Playwright, GitHub Actions CI, Vercel, Render, Git, Linux, Google Cloud Platform

Experience

Open Source Contributor - Agent Orchestrator (Untrivial-ai, 12.6k GitHub stars)

Sep 2026 - Present

Go, TypeScript PR #5638 PR #5956

Remote

- Opened a PR (under review; +577 code and +694 test lines) that fixes issue #3322 by surfacing each coding agent's context-window usage on the session model, CLI and doctor check, so exhausted agents can be respawned instead of nudged.
- Opened a second PR (under review) that treats an update-check timeout as a network failure instead of a crash-level error.

Google Developer Groups VIT-AP - Cloud & DevOps Team

Aug 2024 - Present

Worked with GCP and cloud-native tooling; contributed to developer workshops and community sessions.

Projects

The First Song - Emergent Acoustic Communication between RL Agents | PyTorch, RL

Ongoing

Project page

- Designing and implementing a pre-registered multi-agent RL study (REINFORCE, GAE, PPO from scratch in PyTorch) that tests whether agents sharing one audio channel reorganise their calls in frequency or time, against matched private-channel controls and four baselines.

Hybrid Neural + DTW Retrieval for Cover Song Identification | PyTorch, NumPy, pytest

2026

GitHub

Live

- Cut query time from 2.63 h to 38 ms on the Da-TACOS benchmark by building a two-stage retrieval system: a PyTorch TCN encoder (6 dilated residual blocks, 128-d embeddings, SupCon/triplet loss) shortlists 30 candidates, then a 12-key-rotation subsequence-DTW aligner reranks them.
- Raised hybrid MAP 6.8x (0.018 to 0.122), to within 1.11x of the best system, by running four controlled experiments (hubness correction, 384-frame reranking, training on all 4,780 works, longer training) after the first benchmark showed exhaustive alignment winning 4.6x.

Hulu Support Agent - Evaluated AI Support Agent | Python, Ollama, RAG

2026

GitHub

- Built an AI customer-support agent that classifies each message into 10 intents, drafts a reply grounded in 3 retrieved historical resolutions, and escalates to a human with a stated reason, running fully locally on open-source models (Qwen2.5-7B generator, Llama-3.1-8B judge) served through Ollama.
- Evaluated it on 200 hand-labelled examples against three baselines: macro-F1 0.606 (CI 0.513 to 0.673) vs 0.369 for keyword rules and 0.355 for TF-IDF, with escalation recall (54%) and the weak signal in the model's own confidence (AUC 0.586) reported instead of hidden.

AgentAlign-Lab - Verifier-Guided Preference Learning for LLM Agents | PyTorch, TRL, PEFT, Gradio

2026

GitHub

- Replaced human preference labels with deterministic verifiers for terminal agents by building a ReAct tool-use loop that runs 60 sandboxed tasks across five families (Python bug-fix, config repair, data transformation, log extraction, safety traps) and scores each trajectory on pass/fail, failed commands and unsafe actions.
- Converted scored trajectories into 336 DPO preference pairs and built a QLoRA training path (4-bit NF4 via bitsandbytes, LoRA via PEFT/TRL) for Qwen2.5-Coder-1.5B, with a base-vs-adaptor evaluation on a held-out test split and a Gradio dashboard for inspecting trajectories.

Coil - NSE Swing-Trading Screener and Research Tool | Python, FastAPI, SQLite

2025 - Present

Live

- Built a swing-trading research tool for my own use that scans all about 2,300 listed NSE stocks for Volatility Contraction Patterns plus fundamental and ownership filters, with a FastAPI backend on Render, a JavaScript frontend on Vercel, a nightly GitHub Actions data refresh and 402 passing tests.
- Designed a screen query language (AND/OR/NOT, arithmetic between metrics, Indian number units) with three-valued logic, so a stock with unknown data is shown as unjudged instead of silently failing; one screen found 192 of the 193 stocks on NSE Market Lens, and the live scan and backtest share one strategy state machine.